

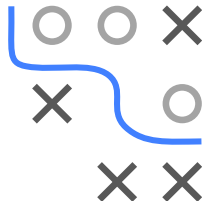
FOLLOW THE REGULARIZED LEADER

- To overcome the shortcomings of the FTL algorithm, one can incorporate a regularization function $\psi : \mathcal{A} \rightarrow \mathbb{R}_+$ into the action choice of FTL, which leads to more stability.
- To be more precise, let for $t \geq 1$

$$a_t^{\text{FTRL}} \in \arg \min_{a \in \mathcal{A}} \left(\psi(a) + \sum_{s=1}^{t-1} \ell(a, z_s) \right),$$

(Technical side note: if there are more than one minimum, then one of them is chosen.)

then the algorithm choosing a_t^{FTRL} in time step t is called the **Follow the regularized leader** (FTRL) algorithm.



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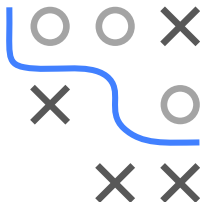
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- *Interpretation:* The algorithm predicts a_t as the element in $\mathcal{A}$, which minimizes the regularization function plus the cumulative loss so far over the previous $t - 1$ time periods.



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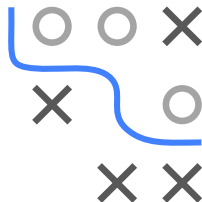
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- *Interpretation:* The algorithm predicts a_t as the element in $\mathcal{A}$, which minimizes the regularization function plus the cumulative loss so far over the previous $t - 1$ time periods.
- Obviously, the behavior of the FTRL algorithm is depending heavily on the choice of the regularization function ψ . If $\psi \equiv 0$, then FTRL equals FTL.

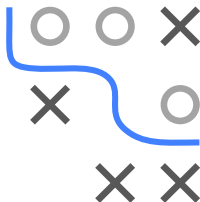


REGULARIZATION IN ONLINE LEARNING VS. BATCH LEARNING

- Note that in the batch learning scenario, the learner seeks to optimize an objective function which is the sum of the training loss and a regularization function:

$$\min_{\theta \in \mathbb{R}^p} \sum_{i=1}^n L(y^{(i)}, \theta) + \lambda \psi(\theta),$$

where $\lambda \geq 0$ is some regularization parameter.



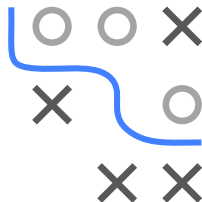
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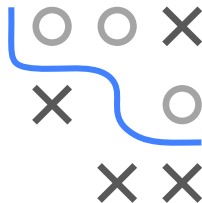
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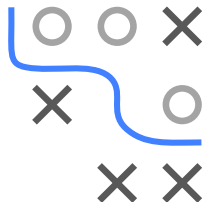
- Here, the regularization function is part of the whole objective function, which the learner seeks to minimize.
- However, in the online learning scenario the regularization function does (usually) not appear in the regret the learner seeks to minimize, but the regularization function is only part of the action/decision rule at each time step.



REGRET ANALYSIS OF FTRL: A HELPFUL LEMMA

- **Lemma:** Let $a_1^{\text{FTRL}}, a_2^{\text{FTRL}}, \dots$ be the sequence of actions coming used by the FTRL algorithm for the environmental data sequence $z_1, z_2, \dots$. Then, for all $\tilde{a} \in \mathcal{A}$ we have

$$\begin{aligned} R_T^{\text{FTRL}}(\tilde{a}) &= \sum_{t=1}^T ((a_t^{\text{FTRL}}, z_t) - (\tilde{a}, z_t)) \\ &\leq \psi(\tilde{a}) - \psi(a_1^{\text{FTRL}}) + \sum_{t=1}^T ((a_t^{\text{FTRL}}, z_t) - (a_{t+1}^{\text{FTRL}}, z_t)) . \end{aligned}$$



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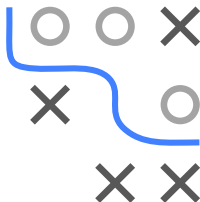
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- *Interpretation:* the regret of the FTRL algorithm is bounded by the difference of cumulated losses of itself compared to its one-step lookahead cheater version and an additional regularization difference term.

⇒ We have seen an analogous result for FTL!

(The proof is similar.)

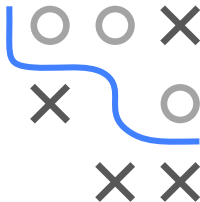


FTRL FOR ONLINE LINEAR OPTIMIZATION

- In the following, we analyze the FTRL algorithm for the linear loss $(a, z) = a^\top z$ for online linear optimization (OLO) problems.
- For this purpose, the squared L2-norm regularization will be used:

$$\psi(a) = \frac{1}{2\eta} \|a\|_2^2 = \frac{a^\top a}{2\eta},$$

where η is some positive scalar, the *regularization magnitude*.



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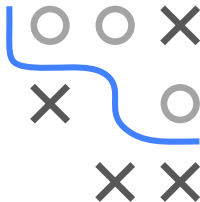
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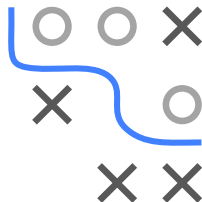
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- Hence, in this case we have for the FTRL algorithm the following update rule

$$a_{t+1}^{\text{FTRL}} = a_t^{\text{FTRL}} - \eta z_t, \quad t = 1, \dots, T-1.$$



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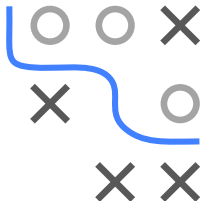
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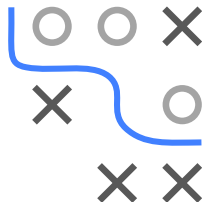
Interpretation: $-z_t$ is the *direction* in which the update of a_t^{FTRL} to a_{t+1}^{FTRL} is conducted with *step size* η in order to reduce the loss.



FTRL FOR OLO: THEORETICAL GUARANTEES

- **Proposition:** Using the FTRL algorithm with the squared L2-norm regularization on any online linear optimization (OLO) problem with $\mathcal{A} \subset \mathbb{R}^d$ leads to a regret of FTRL with respect to any action $\tilde{a} \in \mathcal{A}$ of

$$R_T^{FTRL}(\tilde{a}) \leq \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \eta \sum_{t=1}^T \|z_t\|_2^2.$$



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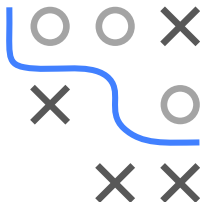
$$R_T^{\text{FTRL}}(\tilde{a}) \leq \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \eta \sum_{t=1}^T \|z_t\|_2^2.$$

- We will show the result only for the case $\mathcal{A} = \mathbb{R}^d$.
- For the more general case, where $\mathcal{A}$ is a strict subset of $\mathbb{R}^d$, we need a slight modification of the update formula above:

$$a_t^{\text{FTRL}} = \Pi_{\mathcal{A}}\left(-\eta \sum_{i=1}^{t-1} z_i\right) = \arg \min_{a \in \mathcal{A}} \left\| a - \eta \sum_{i=1}^{t-1} z_i \right\|_2^2.$$

In words, the action of the FTRL algorithm has to be projected onto the set $\mathcal{A}$. Here, $\Pi_{\mathcal{A}} : \mathbb{R}^d \rightarrow \mathcal{A}$ is the projection onto $\mathcal{A}$.

(The proof is essentially the same, except that the Cauchy-Schwarz inequality is used in between.)



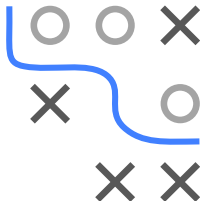
FTRL FOR OLO: THEORETICAL GUARANTEES

- **Proof:**

Reminder (1):
$$R_T^{\text{FTRL}}(\tilde{a}) \leq \psi(\tilde{a}) - \psi(a_1^{\text{FTRL}}) + \sum_{t=1}^T ((a_t^{\text{FTRL}}, z_t) - (a_{t+1}^{\text{FTRL}}, z_t)).$$

Reminder (2):
$$a_{t+1}^{\text{FTRL}} = a_t^{\text{FTRL}} - \eta z_t, \quad t = 1, \dots, T-1.$$

- For sake of brevity, we write $a_1, a_2, \dots$ for $a_1^{\text{FTRL}}, a_2^{\text{FTRL}}, \dots$



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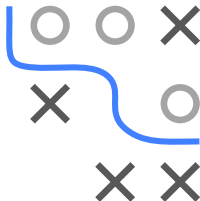
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- With this,

$$\begin{aligned} R_T^{\text{FTRL}}(\tilde{a}) &\leq \psi(\tilde{a}) - \psi(a_1) + \sum_{t=1}^T ((a_t, z_t) - (a_{t+1}, z_t)) && \text{(Reminder (1))} \\ &\leq \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \sum_{t=1}^T (a_t^\top z_t - a_{t+1}^\top z_t) \quad (\psi(a_1) \geq 0 \text{ and definition of } \psi) \\ &= \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \sum_{t=1}^T (a_t^\top - a_{t+1}^\top) z_t && \text{(Distributivity)} \\ &= \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \eta \sum_{t=1}^T \|z_t\|_2^2. && \text{(Reminder (2))} \end{aligned}$$

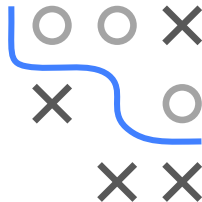
□



FTRL FOR OLO: THEORETICAL GUARANTEES

- Interpretation of the terms in the proposition, i.e., of

$$R_T^{FTRL}(\tilde{a}) \leq \frac{1}{2\eta} \|\tilde{a}\|_2^2 + \eta \sum_{t=1}^T \|z_t\|_2^2 :$$

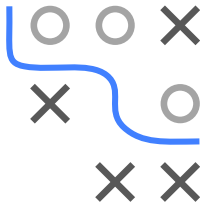


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- $\|\tilde{a}\|_2^2$ represents a *bias term*: The regret upper bound of FTRL is always biased by the term $\|\tilde{a}\|_2^2$. The impact of the bias term can be reduced by a higher regularization magnitude, i.e., a higher choice of η .

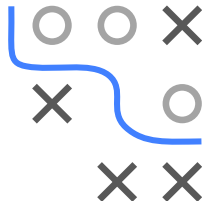


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- $\sum_{t=1}^T \|z_t\|_2^2$ represents a *"variance" term*: The more the environment data z_t varies, the larger this term. Hence, for a high variance a smaller regularization magnitude is needed, i.e., a smaller choice of η .

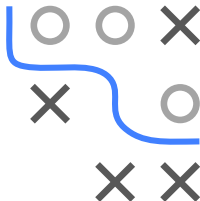


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- Thus, we have a trade-off for the optimal choice of η : Making η large, leads to a smaller *bias* but at the expense of a higher *variance* and making η small leads to a smaller *variance* at the expense of a higher *bias*.

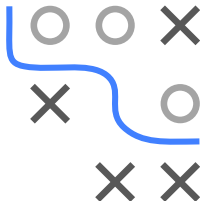


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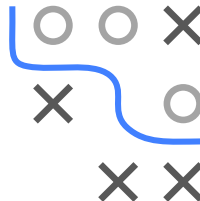
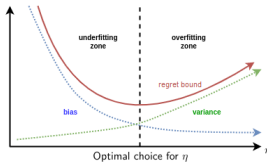
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 - $\sum_{t=1}^T \|z_t\|_2^2$ represents a *"variance" term*: The more the environment data z_t varies, the larger this term. Hence, for a high variance a smaller regularization magnitude is needed, i.e., a smaller choice of η .
 - Thus, we have a trade-off for the optimal choice of η : Making η large, leads to a smaller *bias* but at the expense of a higher *variance* and making η small leads to a smaller *variance* at the expense of a higher *bias*.
- ⇒ With the right choice of η , we can prevent the instability of FTRL for an online linear optimization (OLO) problem.



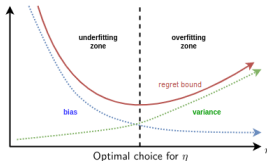
FTRL FOR OLO: THEORETICAL GUARANTEES

- Under certain assumptions we can balance the trade-off induced by the bias and the variance by choosing η appropriately.

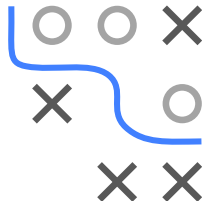


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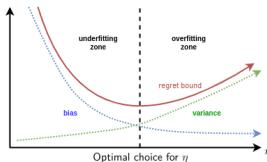


- Corollary:** Suppose we use the FTRL algorithm with the squared L2-norm regularization on an online linear optimization problem with $\mathcal{A} \subset \mathbb{R}^d$ such that
 - $\sup_{\tilde{a} \in \mathcal{A}} \|\tilde{a}\|_2 \leq B$ for some finite constant $B > 0$,
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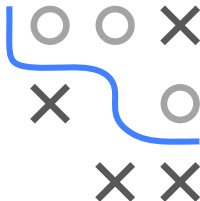


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Then, by choosing the step size η for FTRL as $\eta = \frac{B}{V\sqrt{2T}}$ it holds that

$$R_T^{FTRL} \leq BV\sqrt{2T}.$$



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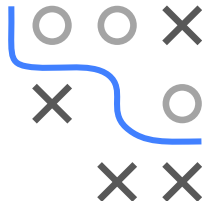
- By the latter **proposition** and the **assumptions**

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$$\leq \frac{B^2}{2\eta}$$

$$+ \eta \sum_{t=1}^T \|z_t\|_2^2$$

$$+ \eta T V^2.$$



FTRL FOR OLO: THEORETICAL GUARANTEES

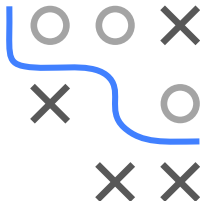
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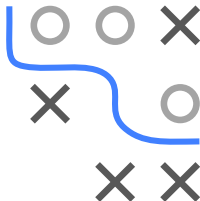
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- Now, the right-hand side of the latter display is a function of the form $f(\eta) = a/\eta + b\eta$ for some suitable $a, b > 0$.



FTRL FOR OLO: THEORETICAL GUARANTEES

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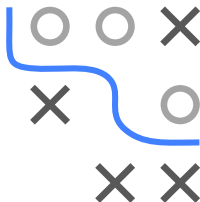
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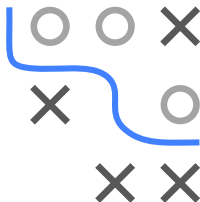
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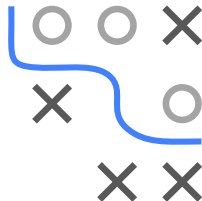
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- Plugging this minimizer into the latter display leads to the asserted inequality. $\square$



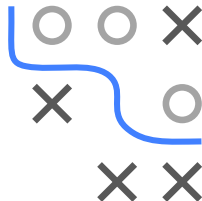
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⇒ But what about other online learning problems or rather other loss functions?

- What we wish to have is an approach such that we can achieve for a large class of loss functions the advantages of FTRL for OLO and OCO problems:
 - (a) reasonable regret upper bounds;
 - (b) a quick update formula.

